

Figure 15-15 shows the resulting chart. Instead of one most likely result, there are five such results. This happens because there's still only one way to roll the smallest value (1 and 1) and the largest value (6 and 10), but the smaller die limits the number of ways you can generate the middle numbers. There are six ways to roll a 7, 8, 9, 10, or 11, these are the most common results, and you're equally likely to roll any one of them.

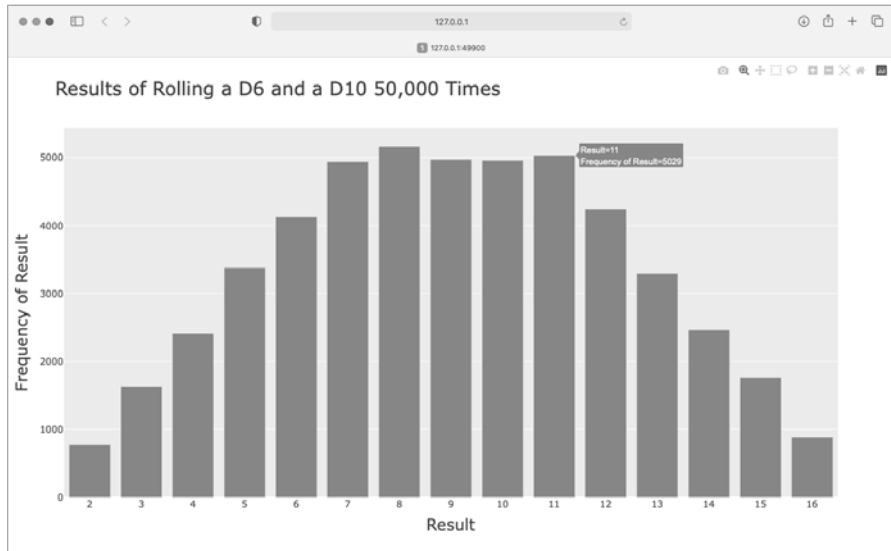


Figure 15-15: The results of rolling a six-sided die and a ten-sided die 50,000 times

Our ability to use Plotly to model the rolling of dice gives us considerable freedom in exploring this phenomenon. In just minutes, you can simulate a tremendous number of rolls using a large variety of dice.

Saving Figures

When you have a figure you like, you can always save the chart as an HTML file through your browser. But you can also do so programmatically. To save your chart as an HTML file, replace the call to `fig.show()` with a call to `fig.write_html()`:

```
fig.write_html('dice_visual_d6d10.html')
```

The `write_html()` method requires one argument: the name of the file to write to. If you only provide a filename, the file will be saved in the same directory as the `.py` file. You can also call `write_html()` with a `Path` object, and write the output file anywhere you want on your system.